

Lecture 1

Functional aspects of learning the case of interpolation

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Appetizer...directional derivative

$\mathcal{H}$ is a RKHS with the dot product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$, and kernel $\kappa(x, y)$. $R_{\lambda} : \mathcal{H} \longrightarrow \mathbb{R}, \lambda \in \mathbb{R}$

$$R_{\lambda}(f) := \sum_{i=1}^m (f(x_i) - y_i)^2 + \lambda \|f\|_{\mathcal{H}}^2 \quad (1)$$

Definition : $\forall f, g \in \mathcal{H}, a \in \mathbb{R}, \left. \frac{d(R_{\lambda}(f + ag))}{da} \right|_{a=0} = \langle R'_{\lambda}(f), g \rangle_{\mathcal{H}},$

Exercise : Compute $R'_{\lambda}(f)$

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$$R_{\lambda}(f) := \sum_{i=1}^m (f(x_i) - y_i)^2 + \lambda \|f\|_{\mathcal{H}}^2 \quad (2)$$

Definition : $\forall f, g \in \mathcal{H}, a \in \mathbb{R}, \left. \frac{d(R_{\lambda}(f + ag))}{da} \right|_{a=0} = \langle R'_{\lambda}(f), g \rangle_{\mathcal{H}},$

Exercise : Compute $R'_{\lambda}(f)$

$$\begin{aligned} \left. \frac{d(R_{\lambda}(f + ag))}{da} \right|_{a=0} &= \left. \frac{d \sum_{i=1}^m (f(x_i) + ag(x_i) - y_i)^2 + \lambda \|f + ag\|_{\mathcal{H}}^2}{da} \right|_{a=0} && \text{using (2)} \\ &= 2 \sum_{i=1}^m (f(x_i) - y_i)g(x_i) + 2\lambda \langle f, g \rangle_{\mathcal{H}} \\ &= 2 \sum_{i=1}^m (f(x_i) - y_i) \langle g, \kappa(\cdot, x_i) \rangle_{\mathcal{H}} + 2\lambda \langle f, g \rangle_{\mathcal{H}} && g(x_i) = \langle g, \kappa(\cdot, x_i) \rangle_{\mathcal{H}} \\ &= 2 \langle g, \underbrace{\sum_{i=1}^m (f(x_i) - y_i) \kappa(\cdot, x_i) + \lambda f}_{R'_{\lambda}(f)/2} \rangle_{\mathcal{H}} && \text{by linearity} \end{aligned}$$

to compute the derivative we need to know how to represent $f(x_i)$

Appetizer...evaluation functional $f(x_i)$

- a real valued **function** $f : \Omega \rightarrow \mathbb{R}$ is not the same than its **value at a given point**

$$\cos(.), \cos(x) \in \mathcal{H} \quad \cos(x_i) \in \mathbb{R}$$

- some functions cannot be evaluated at some points
$$\text{sign}(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

- the evaluation functional $T_{x_i} : \mathcal{H} \longrightarrow \mathbb{R}$

$$f \longmapsto T_{x_i} f = f(x_i)$$

- how to **represent a functional by a function** - 3 ways to evaluate a function at x_i

$$f(x_i) = (\delta_{x_i}, f) = (\delta_{x_i}(.), f(.)) \quad \text{distribution - duality}$$

$$f(x_i) = \int \tilde{f}(\omega) \exp^{i\omega x_i} d\omega \quad \text{Fourier transform}$$

$$f(x_i) = \langle f(.), \kappa(., x_i) \rangle_{\mathcal{H}} \quad \text{reproducing kernel}$$

- define $\mathbb{R}^{\Omega}$ as the set of **pointwise** defined functions from Ω onto $\mathbb{R}$

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to learn we need $\mathcal{H} \subset \mathbb{R}^{\Omega}$

Road map

■ ill posed problem, regularization and Learning

1. splines
2. the interpolation problem
3. well posed and ill posed problem
4. regularization framework
5. Tikhonov regularization
6. other regularization methods
7. regularization analysis through filter functions

■ Iterative algorithm and regularization

■ Semi convergence - the proof

An example of interpolation splines

- inputs : $x \in \mathbb{R}$, output : $y \in \mathbb{R}$
- sample : $S_m = \{x_1, y_1, \dots, x_m, y_m\}$
- find a function f such that $f(x_i) = y_i, i = 1, m$

$$\min_{f \in \mathcal{H}} R_\lambda(f) \quad \text{with } R_\lambda(f) = \sum_{i=1}^m (f(x_i) - y_i)^2 + \frac{1}{\lambda} \int_0^1 (f'(x))^2 dx$$

$$\mathcal{H} = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f' \in L^2[0, 1]\}$$

fit the data

trade off

regularity

$$\begin{aligned} \left. \frac{d(R_\lambda(f + ag))}{da} \right|_{a=0} &= 2 \sum_{i=1}^m (f(x_i) - y_i) g(x_i) + \frac{2}{\lambda} \int_0^1 f'(x) g'(x) dx \\ &= 2 \sum_{i=1}^m (f(x_i) - y_i) (\delta_{x_i}, g) + \frac{2}{\lambda} \left(f'(0) g(0) + f'(1) g(1) - \int_0^1 f''(x) g(x) dx \right) \end{aligned}$$

$$\forall g \in \mathcal{H} \quad \left. \frac{d(R_\lambda(f + ag))}{da} \right|_{a=0} = 0 \quad \Leftrightarrow \quad \sum_{i=1}^m (f(x_i) - y_i) \delta_{x_i} + \frac{1}{\lambda} (f'(0) \delta_0 + f'(1) \delta_1 - f''(x)) = 0$$

An example of interpolation splines

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$$\mathcal{H} = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f' \in L^2[0, 1]\}$$

$$R'_\lambda(f) = 0 \Leftrightarrow \sum_{i=1}^m (f(x_i) - y_i) \delta_{x_i} + \frac{1}{\lambda} (f'(0) \delta_0 + f'(1) \delta_1 - f''(x)) = 0$$

we need two more conditions $f'(0) = 0, f'(1) = 0$

$$\Leftrightarrow f''(x) = \sum_{i=1}^m \alpha_i \delta_{x_i} \quad \text{with } \alpha_i = \lambda(f(x_i) - y_i)$$

$$\Leftrightarrow f'(x) = \sum_{i=1}^m \alpha_i Y(x - x_i) + a \quad \text{but } f'(0) = 0 \Rightarrow a = 0$$

$$\Leftrightarrow f(x) = \sum_{i=1}^m \alpha_i (x - x_i)_+ + b \quad \text{with } f'(1) = 0 \Rightarrow \sum_{i=1}^m \alpha_i = 0$$

$f \in \mathcal{H}$ but f is NOT differentiable... something went wrong - it is the definition of $\mathcal{H}$

The correct functional setup

- the initial idea - select regular function

$$\mathcal{H} = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f' \in L^2[0, 1]\}$$

- Let's do it the other way round - **build** regular functions

$$\mathcal{H} = \left\{ f : \mathbb{R} \rightarrow \mathbb{R} \mid g \in L^2[0, 1], f(x) = \int_0^x g(t) dt \right\}$$

- wrong way

$$\begin{aligned} P : \mathcal{H} &\longrightarrow L^2[0, 1] \\ f &\longmapsto Pf = f' \end{aligned}$$

- pointwise definition

$$\begin{aligned} S_x : L^2[0, 1] &\longrightarrow \mathbb{R} \\ g &\longmapsto S_x g = \int_0^x g(t) dt \end{aligned}$$

- functional setting

$$\begin{aligned} S : L^2[0, 1] &\longrightarrow \mathbb{R}^{[0, 1]} \\ g &\longmapsto f = Sg \text{ s.t. } \forall x \in [0, 1], f(x) = S_x g \end{aligned}$$

- $\mathcal{H} := \text{Im}(S) \subset \mathbb{R}^{[0, 1]}$, S is one to one on $\mathcal{H}$

- $\mathcal{H}$ is a Hilbert space with $\forall f_1, f_2 \in \mathcal{H}, \quad \langle f_1, f_2 \rangle_{\mathcal{H}} := \langle S^{-1} f_1, S^{-1} f_2 \rangle_{L^2[0, 1]}$

The correct functional setup

$$\mathcal{H} = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f' \in L^2[0, 1]\}$$

- general derivatives

$$\mathcal{H} = \left\{ f : \mathbb{R} \rightarrow \mathbb{R} \mid g \in L^2[0, 1], f(x) = \int_0^x g(t) dt \right\}$$

$$S : L^2[0, 1] \longrightarrow \mathcal{H} \subset \mathbb{R}^{[0,1]}$$

$$g \longmapsto f = Sg \text{ such that } f(x) = S_x g$$

- $\mathcal{H}$ is a Hilbert space with $\forall f_1, f_2 \in \mathcal{H}, \quad \langle f_1, f_2 \rangle_{\mathcal{H}} := \langle S^{-1} f_1, S^{-1} f_2 \rangle_{L^2[0,1]}$

- The reproducing property

$$\begin{aligned} \forall f \in \mathcal{H}, x \in [0, 1] \quad f(x) &= \int_0^1 Y(x-t) g(t) dt && (g = S^{-1} f \in L^2[0, 1]) \\ &= \langle Y(x - \cdot), g(\cdot) \rangle_{L^2[0,1]} \\ &= \langle S_{(\cdot)} Y(x-t), S_{(\cdot)} g(t) \rangle_{\mathcal{H}} \\ &= \langle S_{(\cdot)} Y(x-t), f(\cdot) \rangle_{\mathcal{H}} && = \langle \kappa(x, \cdot), f(\cdot) \rangle_{\mathcal{H}} \end{aligned}$$

- $\mathcal{H}$ is a reproducing kernel Hilbert space with kernel

$$\kappa(x, x') = S_{x'} Y(x - \cdot) = \int_0^{x'} Y(x-t) dt = x' - (x' - x)_+$$

Summarize the spline example

■ first way - specify the regularity

■ $\min_{f \in \mathcal{H}} R_\lambda(f)$ with $R_\lambda(f) = \sum_{i=1}^m (f(x_i) - y_i)^2 + \frac{1}{\lambda} \int_0^1 (f'(x))^2 dx$

■ define $\mathcal{H}$

■ the solution is not differentiable !

■ Alternative way - begin by building $\mathcal{H}$

■ choose a family of L^2 functions ($\Gamma_x \in L^2$)

e.g. $\Gamma_x = Y(x - t)$

■ build the associated map S , make it bijective

■ define $\mathcal{H} = \text{Im}(S)$

■ by construction $\mathcal{H}$ is a r.k.h.s. with kernel $\kappa(x, x') = \langle \Gamma_x, \Gamma_{x'} \rangle_2$

■ $\min_{f \in \mathcal{H}} R_\lambda(f)$ with $R_\lambda(f) = \sum_{i=1}^m (f(x_i) - y_i)^2 + \frac{1}{\lambda} \|f\|_{\mathcal{H}}^2$

■ the solution is $f(x) = \sum_{i=1}^m \alpha_i \kappa(x, x_i) + \sum_{j=1}^p b_j \varphi_j(x)$

e.g. $p = 1, \varphi_1(\cdot) = 1$

Summarize the spline example

■ first way - specify the regularity

■ $\min_{f \in \mathcal{H}} R_\lambda(f)$ with $R_\lambda(f) = \sum_{i=1}^m (f(x_i) - y_i)^2 + \frac{1}{\lambda} \int_0^1 (f'(x))^2 dx$

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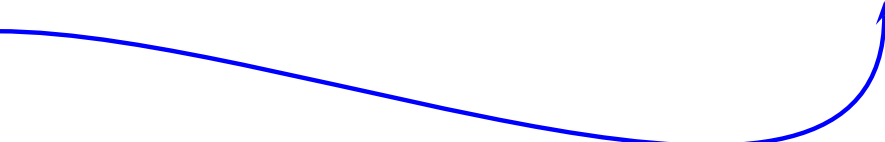
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e.g. $p = 1, \varphi_1(\cdot) = 1$

parametric



The interpolation problem

- inputs : $\mathbf{x} \in \mathbb{R}^d$, output : $y \in \mathbb{R}$
- sample : $S_m = \{\mathbf{x}_1, y_1, \dots, \mathbf{x}_m, y_m\}$

Definition (Evaluation functional)

$$T : \mathcal{H} \longrightarrow \mathbb{R}^m$$

$$f \longmapsto Tf = (f(\mathbf{x}_1), \dots, f(\mathbf{x}_i), \dots, f(\mathbf{x}_m))^\top$$

Definition (Interpolation problem in $\mathcal{H}$)

find $f \in \mathcal{H}$ such that $Tf = \mathbf{y}$

To be well posed the interpolation problems needs

- choose $\mathcal{H}$ such that T is well defined and continuous
- add a constraint to get unicity

Definition (Minimum norm interpolation problem)

Let $\mathcal{H} \subset \mathbb{R}^\Omega$ in which the evaluation functional T is continuous ($\mathcal{H}$ is a r.k. space)

$$\min_{f \in \mathcal{H}} \|f\|^2 \text{ such that } Tf = \mathbf{y}$$

let's solve the minimum norm interpolation problem (M.N.I.P.)

Representer theorem for the M.N.I.P.

Representer theorem

$$\begin{cases} \min_{f \in \mathcal{H}} \|f\|^2 \\ \text{such that } Tf = \mathbf{y} \end{cases} \Leftrightarrow \begin{cases} f = T^* \alpha \\ \text{solve } K\alpha = \mathbf{y} \end{cases} \quad K_{ij} = \kappa(x_i, x_j)$$

evaluation operator and its adjoint

$$\begin{aligned} T : H &\longrightarrow \mathbb{R}^m \\ f &\longmapsto Tf = (f(x_1), \dots, f(x_m))^\top \end{aligned}$$

$$\begin{aligned} T^* : \mathbb{R}^m &\longrightarrow H \\ \alpha &\longmapsto T^* \alpha = \sum_{i=1}^m \alpha_i \kappa(\cdot, x_i) \end{aligned}$$

build the adjoint

$$\begin{aligned} \langle Tf, \alpha \rangle_{\mathbb{R}^m} &= \sum_{i=1}^m f(x_i) \alpha_i \\ &= \sum_{i=1}^m \langle f, \kappa(\cdot, x_i) \rangle_{\mathcal{H}} \alpha_i \\ &= \langle f, \underbrace{\sum_{i=1}^m \alpha_i \kappa(\cdot, x_i)}_{T^* \alpha} \rangle_{\mathcal{H}} \end{aligned}$$

$$\begin{aligned} TT^* : \mathbb{R}^m &\longrightarrow \mathbb{R}^m \\ \alpha &\longmapsto TT^* \alpha = \sum_{j=1}^m \alpha_j \kappa(x_j, x_i) \\ &= K\alpha \end{aligned}$$

the eigen structure. Assume K is full rank - $K\phi_i = \mu_i \phi_i$

the use of the eigen structure

- Assume K is full rank - $K\phi_i = \mu_i\phi_i$
- functions $T^*\phi_i, i = 1, m$ generates F_m a linear subspace of $\mathcal{H}$, such that $\mathcal{H} = F_m \oplus G$ note that $\forall g \in G, Tg = 0$

$$\begin{aligned}\forall f \in \mathcal{H}, \quad f(.) &= \sum_{i=1}^m \beta_i T^* \phi_i + g(.) \\ &= \sum_{i=1}^m \beta_i \left(\sum_{j=1}^m \phi_{ij} \kappa(., x_j) \right) + g(.) \\ &= \sum_{j=1}^m \left(\sum_{i=1}^m \beta_i \phi_{ij} \right) \kappa(., x_j) + g(.) \\ &= \sum_{j=1}^m \alpha_j \kappa(., x_j) + g(.) = T^* \alpha + g(.)\end{aligned}$$

- $Tf = \mathbf{y} \Leftrightarrow TT^* \alpha = K\alpha = \mathbf{y}$
- because we are looking for the minimum norm solution, $g = 0$

summarize the M.N.I.P.

$$\begin{cases} \min_{f \in \mathcal{H}} \|f\|^2 \\ \text{such that } Tf = y \end{cases}$$

- chose the kernel $\kappa(.,.)$
- build the gram matrix K , $(K_{ij} = \kappa(x_i, x_j))$

■ solve $K\alpha = y$

■ $f = T^*\alpha = \sum_{j=1}^m \alpha_j \kappa(., x_j)$

■ cost $\mathcal{O}(m^3)$

■ what if K is singular?

regularize more

■ what is we want to approximate instead of interpolate?

regularize more

■ what if there is no norm?

use orthogonality

■ project 0 onto the interpolating set $I_{\mathcal{H}} = \{f \in \mathcal{H} \text{ such that } Tf = y\}$,

■ project $I_{\mathcal{H}}$ onto $F_m = \text{span}\{\kappa(., x_1), \dots, \kappa(., x_m)\}$

Conclusion

- interpolation from data is a function retrieval problem
- the choice of the hypothesis space $\mathcal{H}$ is very important step
 - specify what is a "good" solution
 - $\mathcal{H}$ has to be a set of *pointwise defined* function
 - it's good if it has the *reproducing property*
- choosing $\mathcal{H}$ can be seen as the choice of the kernel
- move from function to m dimensional vector : *reproducing theorem*
- regularization will come in a second step : lecture 2

3 ways to specify $\mathcal{H}$: *regularity*, *kernel* or the *S operator*